

Tu-154M

Aerodynamic Analysis & Flap/Stabilizer Coupling

English translation of the two supplied technical analyses

Part I: Full Analysis of Cycle Dump(2)

Part II: v4.2.6 Flap/Stabilizer Coupling Calculation

Technical values, units, airfoil names, selector designations, and the original numerical formatting are preserved.

Part I - Tu-154M: Full Analysis of Cycle Dump(2)

1. Identification of the Runtime Airfoil Set

All 70 surface elements and 93 aerodynamic summary blocks were read.

The five runtime names are:

- wing.afl
- wing_tip.afl
- hstab.afl
- vstab.afl
- NACA 0009 (symmetrical).afl

Mean/maximum absolute lookup deviation from the installed correct set; the remaining differences are caused by the lookup-alpha value being rounded to 0,1° in the dump:

- wing.afl: CL 0,001942/0,004360; CD 0,000030/0,000050;
Cm 0,000390/0,000900
- wing_tip.afl: CL 0,002512/0,004343; CD 0,000029/0,000049;
Cm 0,000341/0,000616
- hstab.afl: CL 0,000295/0,000410; CD 0,000032/0,000040;
Cm 0,000012/0,000024
- vstab.afl: CL 0,000100/0,000100; CD 0,000030/0,000040;
Cm 0,000000/0,000000
- NACA file: CL 0,000000/0,000000;
CD 0,000025/0,000034; Cm 0,000000/0,000000

2. Flight Condition

- Time 597,53 s
- Altitude 4.204 m; 175,3 m/s; Mach 0,5415
- alpha 3,2°; pitch attitude 2,5°; pitch rate 0,0°/s
- Mass 80.129,5 kg; load factor 0,9999 g
- CG -0,22 m from the reference point
- Pitch input 0,0000
- Landing gear/flaps retracted; additional stabilizer angle 0,0°
- Engine thrust 11,0 kN per engine
- Elevator -1,4°; dump convention positive down, therefore 1,4° nose-up

3. Surface Groups

Group / count / area / lift / drag / direct moment /
contribution to aircraft pitching moment:

- Wing 1: 20 / 104,54 m² / +431,8 kN / +14,5 kN / -1.359,3 kN m / +145,3 kN m
- Wing 2: 16 / 69,72 m² / +224,0 kN / +7,1 kN / -330,3 kN m / -650,1 kN m
- Wing 3: 16 / 27,92 m² / +78,1 kN / +2,6 kN / -48,0 kN m / -522,5 kN m
- Wing 4: 2 / 0,78 m² / +0,6 kN / +0,1 kN / -0,4 kN m / -5,2 kN m
- Horizontal tailplane: 6 / 37,28 m² / -49,8 kN / +2,3 kN / +40,4 kN m / +1.005,6 kN m
- Vertical tail: 6 / 28,15 m² / 0,0 kN / +1,2 kN / 0,0 kN m / +4,0 kN m
- Other wings: 4 / 2,45 m² / 0,0 kN / +0,2 kN / 0,0 kN m / -0,3 kN m

The zero forces of the vertical tail and the other wings at beta 0° confirm the intended symmetry of the set used.

4. Pitching-Moment Balance

- after all main wings: -1.032,5 kN m
- horizontal-tail contribution: +1.005,6 kN m
- after horizontal tail: -26,8 kN m
- final aerodynamic moment: +264,1 kN m
- final engine moment: -9,3 kN m
- final total moment before asymmetric loading: +0,356 kN m

The run is therefore practically exactly in moment equilibrium.

5. Horizontal Tailplane

- Downwash 1,2 to 1,8°
- Wing-plane alpha 1,8 to 2,6°
- geometric alpha -1,2 to -0,4°
- final CL -0,1938 to -0,1243
- total downforce -49,8 kN
- Elevator -1,4° nose-up

The elevator is within the desired normal range of 0 to 10° nose-up and has very large reserve in both directions relative to +25° up/-20° down.

6. Stability

For a 1° pitch offset, X-Plane determines:

- Moment -9.207 N m/degree
- Cm-alpha -0,0007/degree or -0,0414/rad

The negative sign indicates static longitudinal stability at the test point.

Part II - Tu-154M v4.2.6: Flap/Stabilizer Coupling Calculation

Basis

- correct v4.2.3 AFL tables, confirmed by Cycle Dump(2)
- ACF: Flap 1 on Wing-1 elements 2-7; Flap 2 on all eight Wing-2 elements
- Reference MAC 17,34 ft
- Stabilizer: F 0/3/5,5°, C 0/1,5/3°, A 0/0/0°
- Elevator limits +25° up / -20° down
- NACA-0009 horizontal tailplane and actual tailplane area/lever arm

Clean Anchor from Cycle Dump(2)

Mach 0,5415, alpha 3,2°, additional stabilizer angle 0°, elevator 1,4° nose-up, pitch rate 0,0°/s and final residual moment +356,3 N m. Because the flap deflection is zero in the clean configuration, the new C_m characteristic does not change this condition.

Calculation Method

For each flap, the local CL force arm and the direct airfoil moment are normalized to $S \cdot MAC$ over the ACF elements that are actually equipped with flaps. The two flap moments are scaled by their respective actual deflections. The stabilizer contribution is determined from the NACA-0009 polar, tail volume, and finite span. For each detent, the C_m factor that would exactly neutralize selector C is determined first. 80 percent of the difference from the previous factor is used in order to retain a small stable residual effect.

Exact Neutral Factors for C

- 15°/13°, stabilizer 1,5°: 1,6420
- 28°/25°, stabilizer 1,5°: 1,2238
- 36°/32°, stabilizer 3,0°: 1,4821
- 45°/40°, stabilizer 3,0°: 1,3386

Applied 80-Percent Factors

- 15°/13°: 1,5136
- 28°/25°: 1,1790
- 36°/32°: 1,3857
- 45°/40°: 1,2709

Applied to the previous C_m values -0,296/-0,335, these produce the supplied table values. 451 intermediate points from 0 to 45° were checked.

Continuous Coupling Envelope

Selector F:

- additional elevator requirement 0 to 2,92° nose-down

- largest residual Cm +0,05095

Selector C:

- additional elevator requirement 0 to 0,90° nose-down
- largest residual Cm +0,01569

Selector A:

- additional elevator requirement 0 to 3,03° nose-up
- largest residual Cm -0,05311

Detent Values for C

- 0°/0°, stabilizer 0°: 0,00° additional elevator
- 15°/13°, stabilizer 1,5°: 0,49° nose-down
- 28°/25°, stabilizer 1,5°: 0,32° nose-down
- 36°/32°, stabilizer 3,0°: 0,90° nose-down
- 45°/40°, stabilizer 3,0°: 0,79° nose-down

The figures describe only the change caused by the coupled flap/stabilizer movement. Normal flight-condition trim is added on top of this. The clean anchor remains the 1,4° nose-up requirement measured in the dump.